

Alessandro Bruno

Software Engineer — Rust / Backend (event-driven systems)
Bologna, Italy | alessandro.brunoh@gmail.com | github.com/alessandrobrunoh
EU citizen · based in Bologna · remote or hybrid Italy · available with notice

PROFILE

Rust backend engineer, Bologna. **1+ year** on production event-driven microservices (Tokio, Valkey Streams, S3, OpenTelemetry/Grafana). B.Sc. Computer Systems Technologies, UniBo; thesis on an async telemetry platform. Merged upstream contribution to Zed (PR #3339).

EXPERIENCE

Software Developer Sep 2025 – Present
Luna S.r.l. Bologna, Italy
Intern Sep 2025 – Jun 2026 · Employee Jun 2026 – Present

- **1+ year** on production **event-driven Rust** services for client workloads: async consumers on Valkey Streams, S3 payloads, traces/logs via OpenTelemetry (Grafana, Loki, Tempo).
- **Intern:** fleet platform in **Spring Boot, Angular and React Native** (WebSockets). After two months, moved onto the client Rust backend under senior **code review** (stacked Git branches, reviewable diffs).
- **Now:** same Rust platform plus a new operator console in **Spring Boot and React** (APIs + UI).

OPEN SOURCE

Zed — JDL Language Support · Merged upstream, PR #3339
Rust · Tree-sitter · Zed Extensions

- Merged JDL language support for Zed via a Tree-sitter grammar and editor extension, now in the official marketplace.

Zed — New UI & Dynamic Port Forwarding · Open, PR #55248
Rust · SSH · Remote Development

- Open PR: Port Forwarding UI and **live SSH port forwards** on remote sessions, so adding or removing forwards does not require reconnecting.

ducklake-orm
Rust · DuckDB / DuckLake · crates.io

- Published Rust ORM (**crates.io**) with derive-macro type safety, query builders, connection pooling, schema migrations and DuckLake time-travel.

SELECTED PROJECTS

Mnemosyne
Rust · Tree-sitter · redb · Content-Addressable Storage · MCP

- Local-first tool for **semantic source-code history**, tracking symbols and structural changes through Tree-sitter rather than line-based diffs, with a daemon, CLI/TUI and storage workspace.

Eivar Online
Rust · Bevy · SpacetimeDB · WebAssembly · Distributed State

- Multiplayer prototype: **server-authoritative Rust simulation**, replicated state, client-side prediction and a SpacetimeDB module compiled to WebAssembly.

EDUCATION

B.Sc. in Computer Systems Technologies 2026
University of Bologna — Cesena Campus Cesena, Italy

Thesis: Design and Development of PETRA, an Event-Driven Platform for Real-Time Asynchronous Telemetry and Analysis.
Event-driven telemetry pipeline: async ingest, processing, storage and analysis (aligns with the Luna Rust platform).

TECHNICAL SKILLS

Languages	Rust, TypeScript, Java, Python
Backend & Systems	Tokio, Spring Boot, React, PostgreSQL, Valkey/Redis, WebSockets, event-driven architectures
Infrastructure	Docker, S3, Linux, OpenTelemetry, Grafana, Loki, Tempo
Additional	Git (stacked branches), Tree-sitter, WebAssembly, Bevy · LLM-assisted coding with mandatory review

LANGUAGES

Italian: Native · **English:** Professional working proficiency (interviews in English)